

MEWS Development Program

Testing & Validation — Test Case and Verification Sample

Purpose: showcase structured testing across the software workflow, MEWS alerting, user actions, device integration, hardware reliability, battery, firmware, and burn-in activities.

Source basis: supplied Dry Run Test Cases and Band/Gateway Test Requirements. The original materials contain project-specific names and controlled document information; this portfolio artifact is anonymized and does not claim execution results that were not supplied.

1. Representative Test Case Register

ID	Area	Scenario	Preconditions / Test Basis	Expected Result	Status
DR-TC-001	Dashboard	Initial load & high-risk sorting	25 active records; one high-risk case	All active records load and high-risk case is prioritized	Not Executed
DR-TC-002	MEWS Alerting	Critical alert triggering	Simulated MEWS ≥ 5	Critical alert is triggered and prominently displayed	Not Executed
DR-TC-003	User / Audit	Alert acknowledgement & audit	Active unacknowledged critical alert	Acknowledgement and intervention record are timestamped/retrievable	Not Executed
DR-TC-004	Patient	Discharge & archival	Patient ready for discharge	Patient leaves active list and record remains accessible	Not Executed
DR-TC-006	Gateway / Band	Device disconnection	Connected device intentionally disconnected	Offline/disconnected status and technical alert appear	Not Executed
DR-TC-008	Login / User	First-time login	Temporary account credentials	Mandatory password change is enforced	Not Executed
DR-TC-009	System Setup	Configuration verification	Rooms, gateways and bands configured	Configured entities and relationships are correct	Not Executed
HW-FT-001	Band / Gateway	Sensor accuracy	Compare against calibrated equipment	Measurements meet defined acceptance criteria	Not Executed
HW-RT-001	Reliability	Mechanical / physical stress	Drop, vibration, shock, strap/buckle tests	Device remains within defined functional criteria	Not Executed
HW-BAT-001	Battery	Battery life & protection	Charge/discharge and protection checks	Battery behavior meets defined criteria	Not Executed
HW-FW-001	Firmware	Firmware & wireless reliability	Firmware and wireless communication checks	Firmware and communication operate as specified	Not Executed
HW-BI-001	Burn-in	Environmental / burn-in	Temperature, power cycling, waterproofing	No unacceptable failure during defined burn-in	Not Executed

2. Testing & Validation Coverage

Test Layer	Coverage	Evidence / Output
Software / Functional	Dashboard, MEWS scoring, alerting, user actions, discharge, configuration	Test cases, expected results, execution record
Integration	Band/gateway communication, device status, wireless behavior	Integration test evidence and defect follow-up
Hardware Reliability	Accuracy, drop, vibration, shock, strap/buckle durability	Test results against defined criteria
Battery	Life, charging/discharging, overcharge and over-discharge protection	Battery test record
Firmware	Reliability, wireless communication, firmware security	Firmware test results
Burn-in / Environmental	Temperature, functional test, power cycling, waterproofing, failure analysis	Burn-in record and failure analysis
Validation	Testing in defined application / operating conditions	Validation evidence and acceptance conclusion

3. Verification Record Structure

A practical execution record should capture the test case identifier, environment or device under test, prerequisites, steps performed, expected result, actual result, evidence/screenshots where applicable, status, defects or remarks, and reviewer/approval information.

The supplied Dry Run Test Cases already use a structured format containing **Feature, PBI, Test Case ID, Scenario, Persona, Prerequisites, Test Steps, Expected Results, Actual Results, Status, Test Data, Screenshots, and Remarks**. This portfolio sample preserves that testing mindset while removing project-specific identities.

4. Validation Approach

Validation focuses on whether the solution works for its **defined application and intended operating conditions**, rather than only whether individual technical requirements pass. The supplied requirements include functional accuracy checks, continuous/periodic measurement under different conditions during clinical validation, and defined burn-in and reliability activities.

5. PM Contribution

The Technical Project Manager coordinates test scope, priorities, dependencies, environments, responsible teams, execution readiness, defect and issue follow-up, evidence collection, review checkpoints, and release/validation readiness. Engineering teams remain responsible for the detailed technical execution and test results.

6. Evidence Boundary

This sample intentionally contains no patient-identifying information, credentials, proprietary algorithms, supplier confidential data, controlled test records, or fabricated pass/fail results. All statuses shown above are **Not Executed** because the supplied source material did not provide execution results for these representative cases.